WEKA Explorer is a graphical interface used to perform data preprocessing, classification, clustering, and visualization easily. It is part of the WEKA data mining tool developed by the **University of Waikato**, and we select *Explorer* to build and test machine learning models.

Fig-2: Weka Explorer Interface

This image displays the preprocessing stage where text data is converted into numerical features using the StringToWordVector filter. The dataset contains many attributes generated from words and the class label distribution (fake and real).

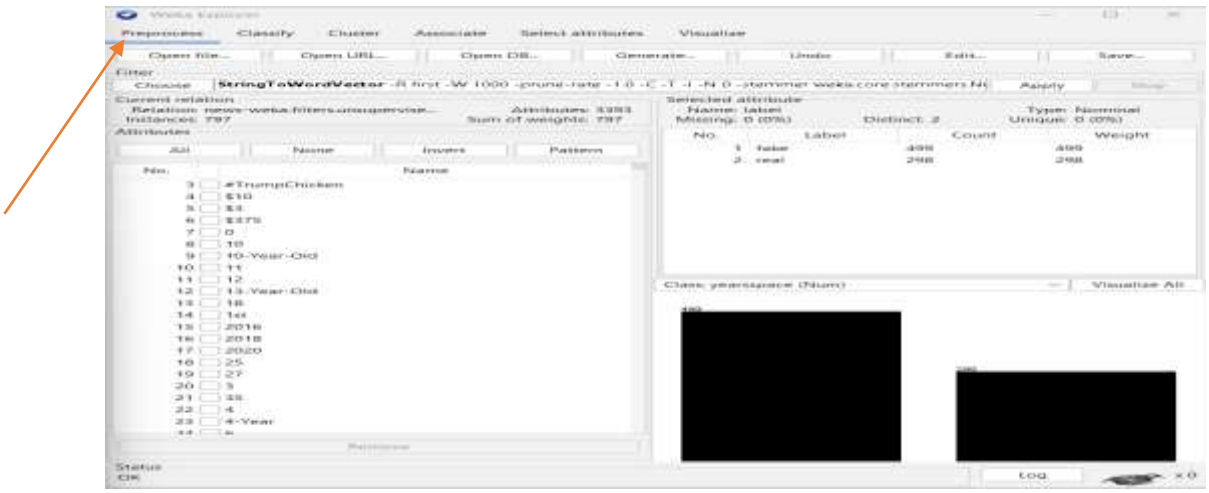
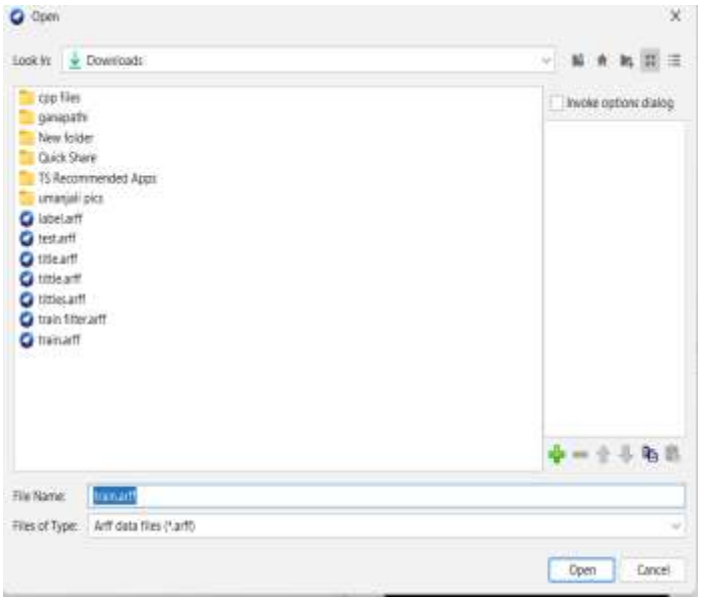


Fig-3: Selecting Preprocess tab



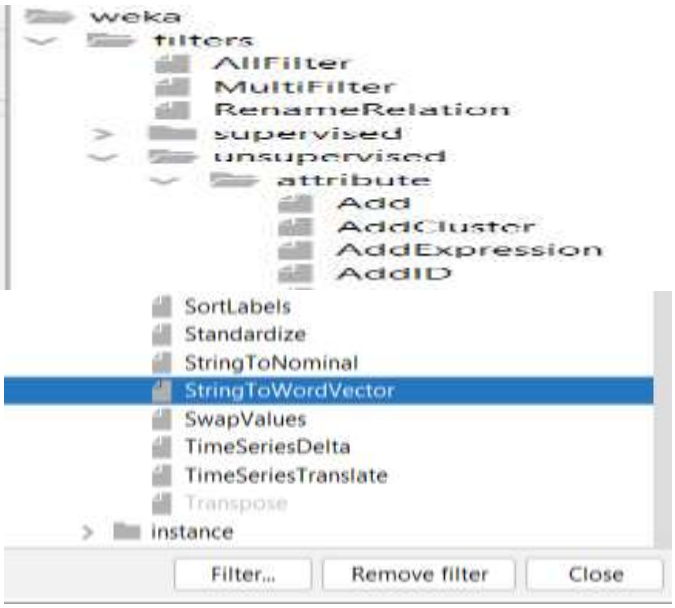


fig-4: Opening dataset file in weka

fig-5:selecting filter in preprocess tab

TF and IDF conversion is used to transform text into numerical values so machine learning algorithms can understand the importance of each word. This conversion highlights important words and reduces the impact of common words, improving classification accuracy.

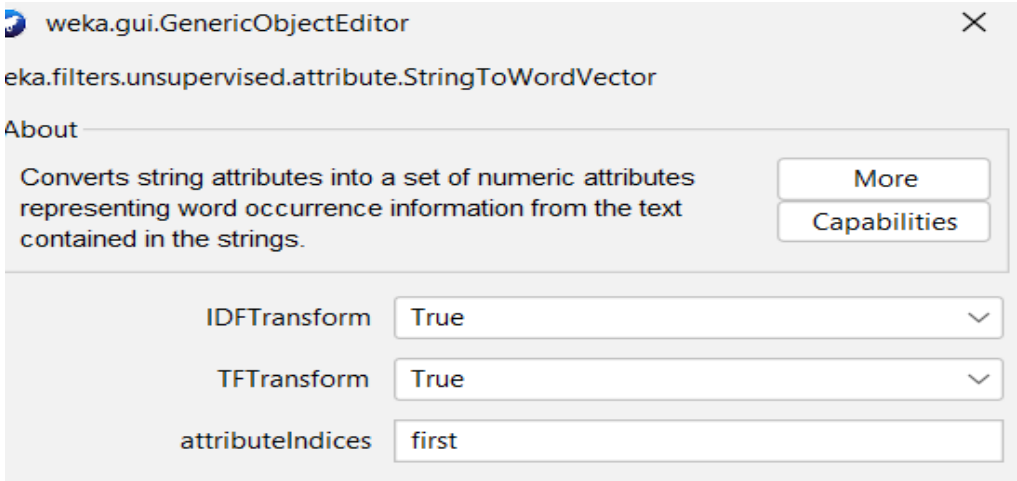


fig-6: TF-IDF conversion

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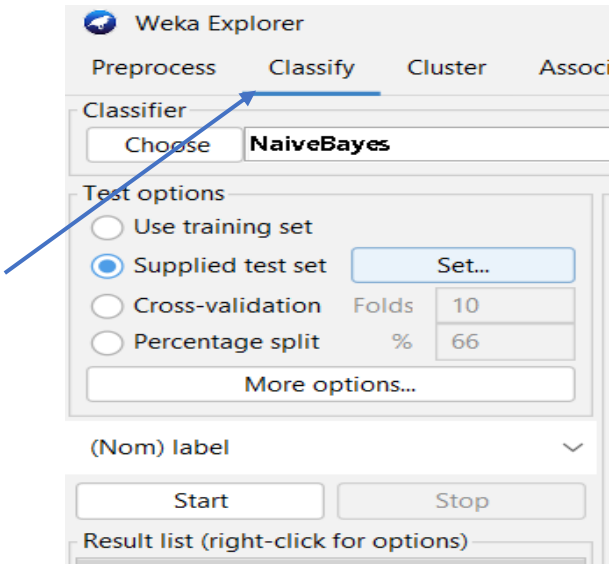
The Classify tab is used to train and evaluate machine learning models on the dataset. Naive Bayes is selected because it is simple, fast, and effective for text classification tasks like fake news detection.



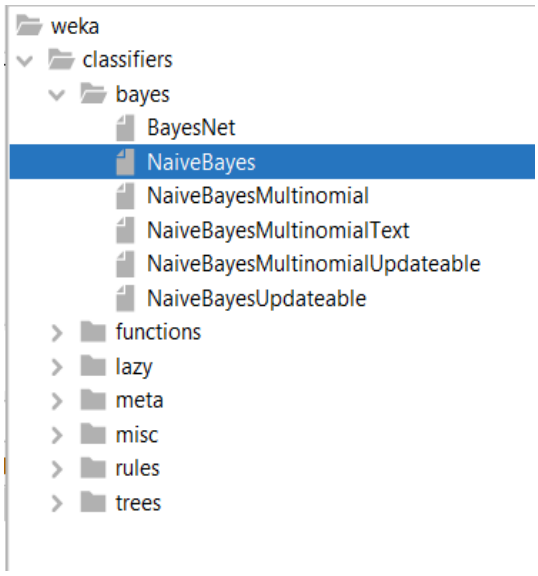


fig-7: Selecting classify tab

fig-8: Choosing Naïve Bayes

This screen shows the Naive Bayes classifier results using cross-validation on the training dataset.The model achieved high accuracy with detailed metrics like precision, recall, and confusion matrix.

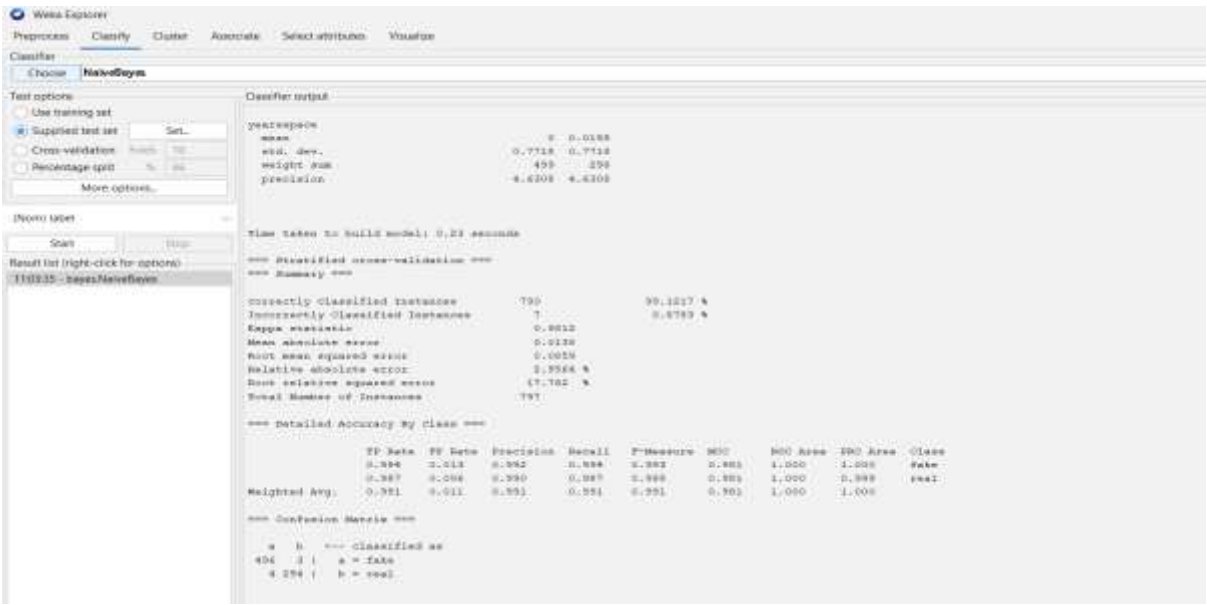


Fig-10: Accuracy result for Training data

This image shows Naive Bayes evaluation using percentage split for testing the model. The classifier achieved high classification accuracy with minimal errors on the test data.

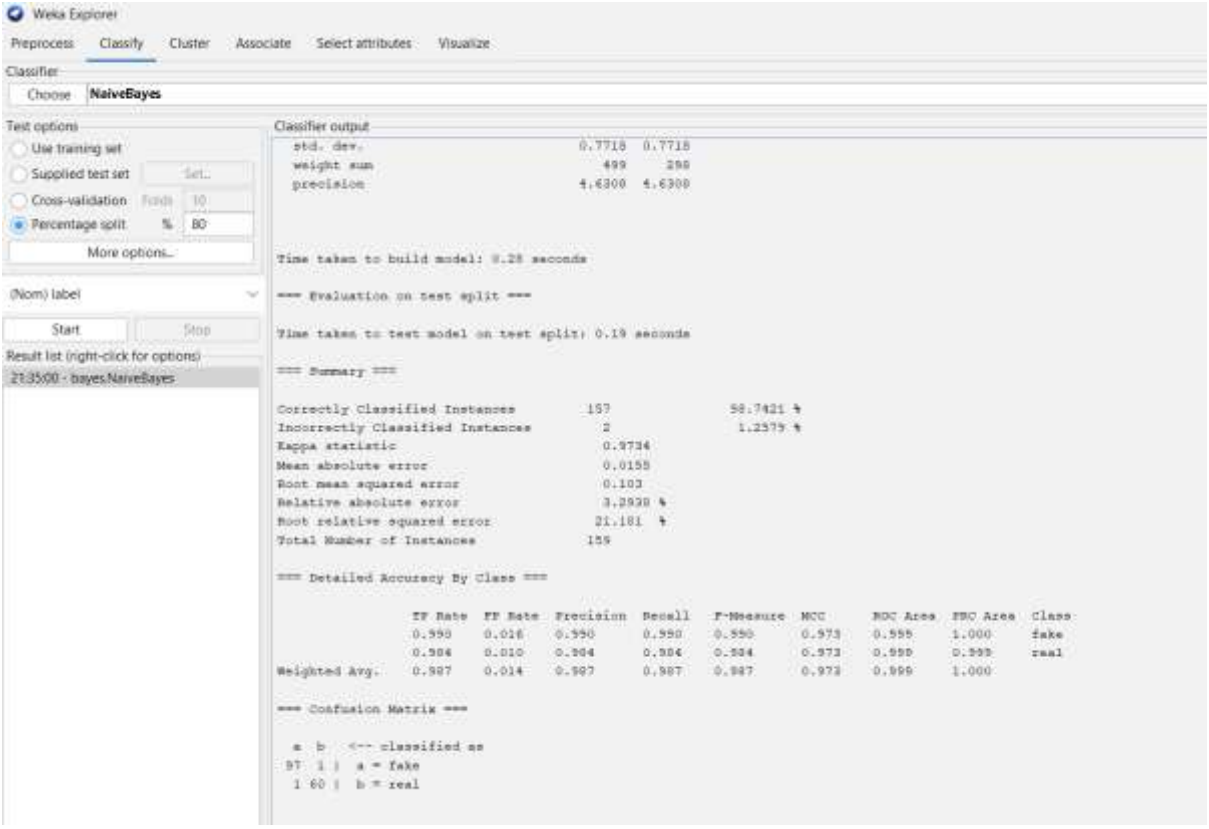


fig-11: Accuracy result for testing data